

RAFFLES INSTITUTION

2011 Year 6 Preliminary Examination

Higher 1

BIOLOGY

8875/01

Paper 1 Multiple Choice

22 September 2011

1 hour

Additional materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write using a soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name and shade your Index Number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions in this paper. Answer all questions. For each question there are four possible answers **A, B, C, and D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Calculators may be used

(Erase all mistakes completely. Do not bend or fold the OMR Answer Sheet).

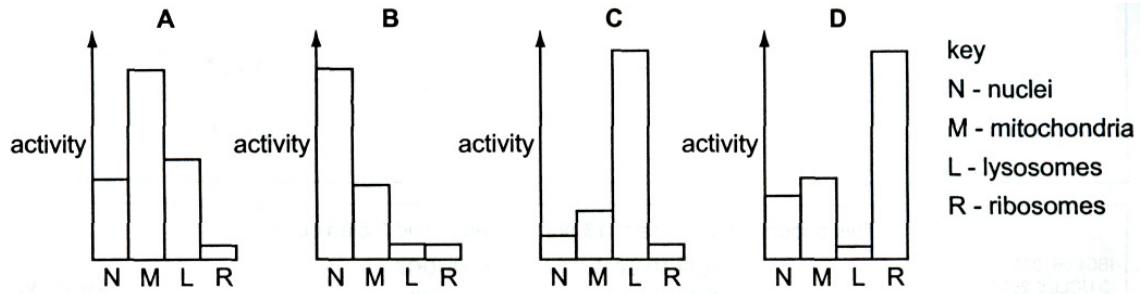
This document consists of **15** printed pages.



Raffles Institution
Internal Examination

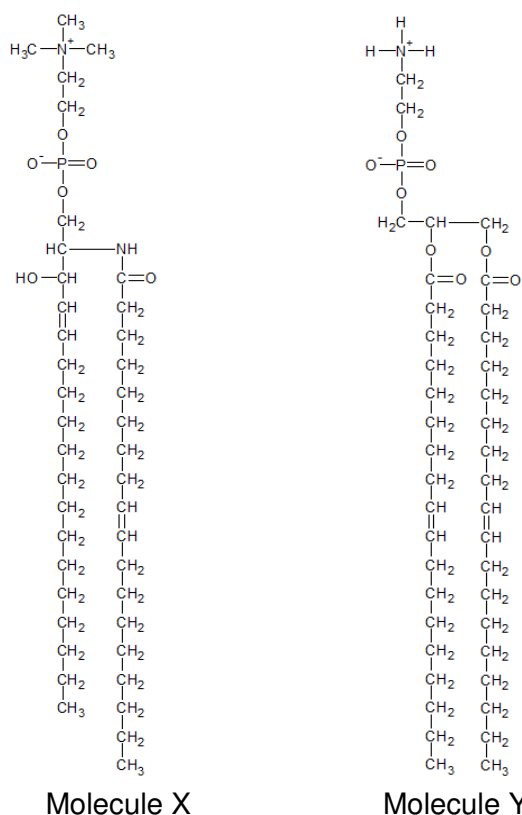
1. A piece of mammalian tissue was homogenised and subjected to differential centrifugation to yield four subcellular fractions. The activity within each fraction, of four different types of enzyme, **A**, **B**, **C** and **D**, was investigated.

Which bar chart shows the results of investigating hydrolytic enzyme activity?



2. From which cell organelle are nucleic acids absent?
- A** Ribosome
 - B** Chloroplast
 - C** Mitochondrion
 - D** Golgi apparatus
3. At which levels of protein structure do hydrophobic interactions occur?
- A** quaternary only
 - B** tertiary and quaternary
 - C** primary, secondary and tertiary
 - D** primary, secondary, tertiary and quaternary

4. The figure shows two biological molecules.

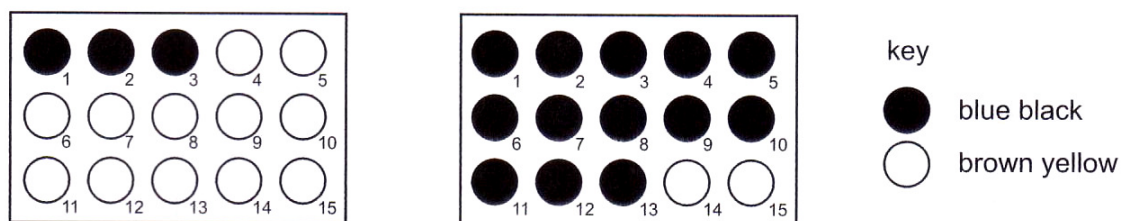


Which of the following statements about the molecules **X** and **Y** is/are true?

I	Only molecule Y is a phospholipid because it does not contain a glycerol backbone.
II	Both molecules X and Y are amphipathic as they both have polar and non-polar components.
III	When molecules X and Y are treated with phospholipases, the same products are produced.

- A** I only
- B** II only
- C** I and III only
- D** II and III only

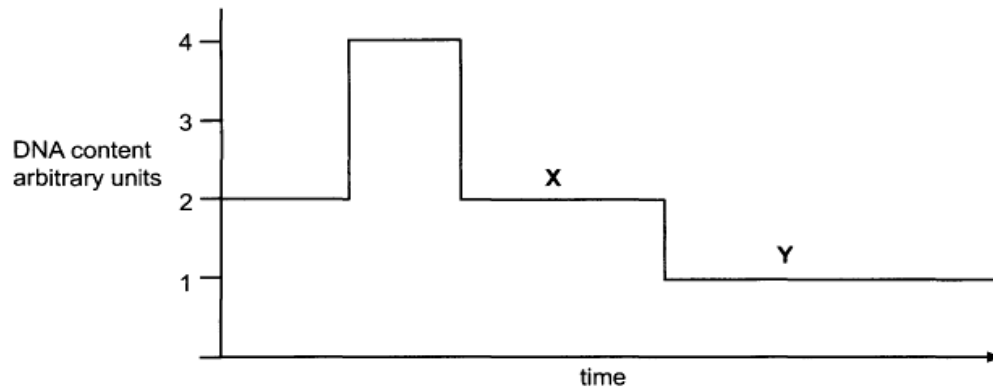
5. An experiment was carried out to investigate the digestion of starch using amylase at two different temperatures. A sample was removed from each mixture at 15 second intervals and placed onto a spotting tile well containing two drops of iodine in KI solution. The results are shown in the diagram.



Which shows the correct temperatures and times for the complete digestion of starch?

	temperature / °C	time / s
A	30	3.15
	10	0.45
B	30	45
	10	195
C	10	45
	30	195
D	10	3.15
	30	0.45

6. The graph shows the amount of DNA present in the nuclei of cells undergoing division in a multicellular organism.

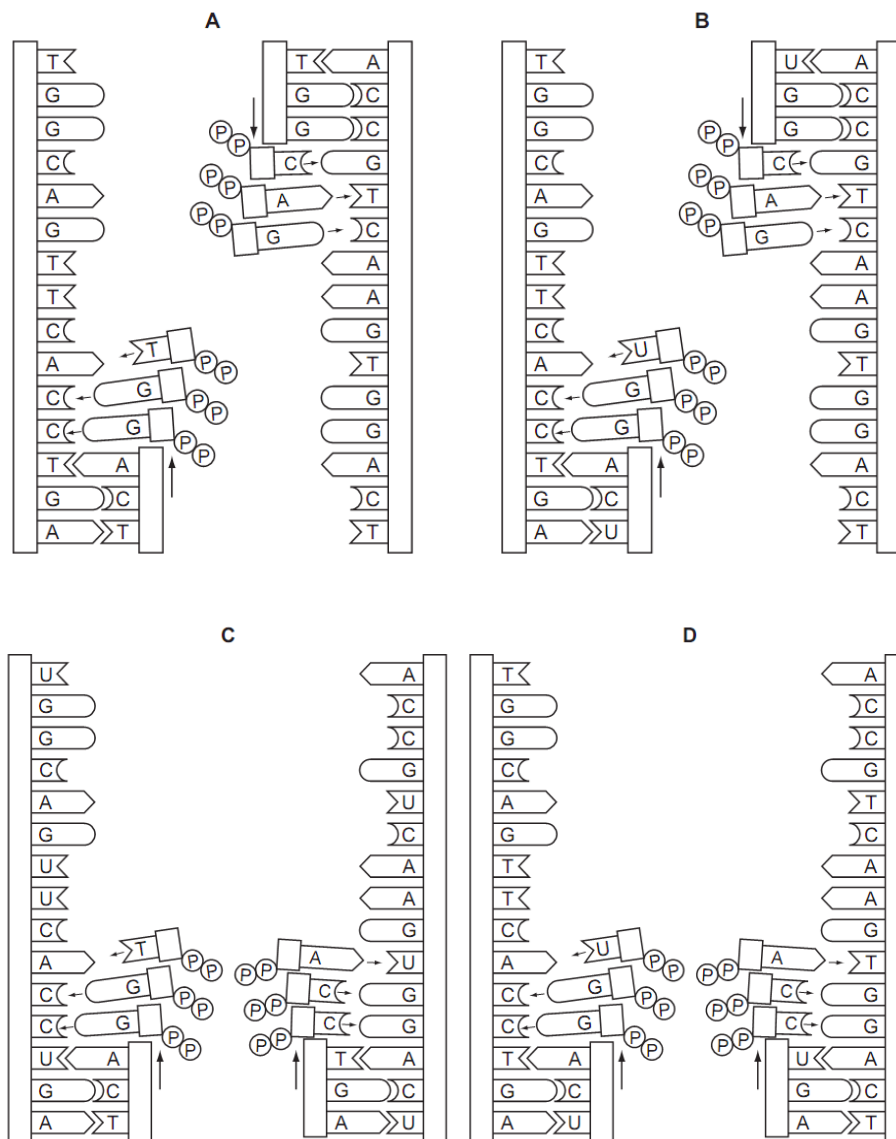


Which combination is correct?

	Ploidy of cell at X	Ploidy of cell at Y
A	diploid	diploid
B	diploid	haploid
C	haploid	haploid
D	haploid	diploid

7. If a proto-oncogene mutated into an oncogene in an otherwise normal cell, the immediate result would likely be
- A** apoptosis.
 - B** arrest of the cell cycle.
 - C** a quickening of the cell cycle.
 - D** rapid mutation of other genes.

8. Which diagram shows the semi-conservative replication of a section of a molecule of DNA?



9. A transcription unit 8000 nucleotides long may use 1200 nucleotides to make a protein consisting of 400 amino acids. This is best explained by the fact that
- A many noncoding nucleotides are present in mature mRNA.
 - B many nucleotides are needed to code for each amino acid.
 - C nucleotides break off and are lost during transcription process.
 - D there are introns in the gene that are the noncoding sequences.

10. In the DNA sequence for sickle cell anaemia, adenine replaces thymine in a CTT triplet, forming the triplet CAT. During synthesis of the sickle cell haemoglobin molecule, the amino acid valine is incorporated instead of glutamic acid.

What is the anticodon in the transfer RNA molecule carrying this valine?

- A CAU
- B CUA
- C GAU
- D GUA

11. The table describes different kinds of mutations in DNA.

mutation	name
from purine to other purine	transition
from pyrimidine to other pyrimidine	transition
from purine to pyrimidine	transversion
from pyrimidine to purine	transversion

The diagram represents part of a DNA molecule.

G A T A C C A
C T A T G G T

Which diagram shows the DNA molecule with only transversion(s)?

A

G T T A T C A
C A A T A G T

B

G A A A C A A
C T T T G T T

C

A A T A C C A
T T A T G G T

D

G A T A T C A
C T A T A G T

12. Some antibacterial drugs can affect the synthesis of proteins.

Antimicrobial drug	rifampicin	streptomycin	tetracycline
Mode of action	Binds to RNA polymerase	Genetic code misread during translation	Prevents binding of tRNA to ribosome

Which is the correct set of immediate effects of these drugs?

Antimicrobial drug	rifampicin	streptomycin	tetracycline
A	Defective protein synthesized	mRNA does not bind to ribosome	Amino acids not added to growing chain
B	Transcription prevented	Defective protein synthesized	Amino acids not added to growing chain
C	Defective protein synthesized	mRNA does not bind to ribosome	Transcription prevented
D	Transcription prevented	Defective protein synthesized	mRNA does not bind to ribosome

13. A man and a woman, both with normal colour vision, have a colour-blind boy together. The woman is pregnant for a second time, and the doctor tells her she is carrying twins of one boy and one girl. What is the chance that both twins will have normal colour vision?

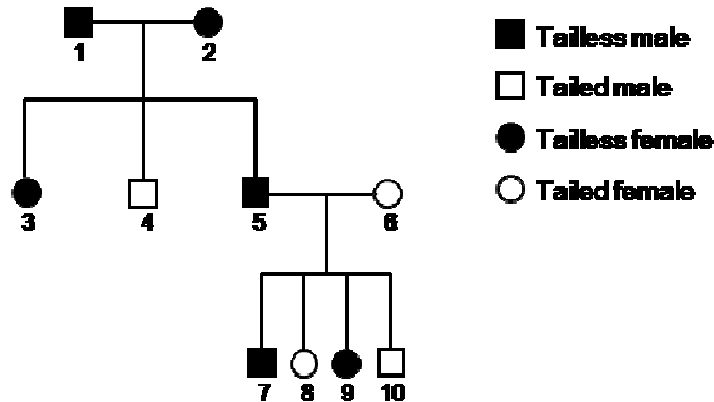
- A** 0
B $\frac{1}{4}$
C $\frac{1}{2}$
D $\frac{3}{4}$

14. A man was brought to court in a paternity case. He has blood type B, Rh positive. The mother has blood type B, Rh negative.

What blood type of the child would exclude the accused from paternity?

- A** AB, Rh negative
B B, Rh negative
C B, Rh positive
D O, Rh negative

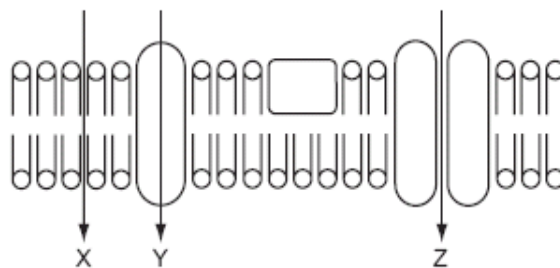
15. The Manx gene determines the presence or absence of tails in cats. Individuals that do not possess the dominant allele develop tails. The pedigree chart below shows the absence or presence of tails in three generations of cats.



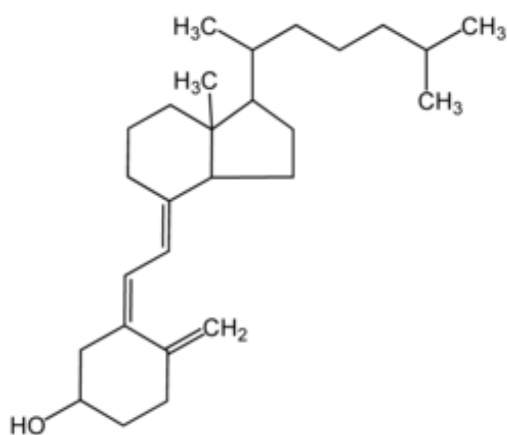
- Which one of the following statements is likely to be true?
- A Individual 2 is homozygous dominant for the Manx gene.
 - B If individual 4 mated with another tailed cat, they would produce tailless offspring.
 - C The homozygous dominant condition is lethal.
 - D The Manx gene is located on the sex chromosome.
16. Aberrations during crossing over in Prophase I of meiosis may result in chromosomal mutations. Which of the following mutations is least likely to result?
- A Deletion
 - B Duplication
 - C Polyploidy
 - D Translocation
17. Which of the following statements regarding glycolysis is *false*?
- A All cells can perform glycolysis.
 - B The substrates for glycolysis include glucose, ADP and NAD^+ .
 - C Glycolysis can occur in either the presence or absence of O_2 .
 - D For each molecule of glucose undergoing glycolysis, the cell gains 4 ATP, 2 NADH + H^+ and 2 pyruvate molecules.

18. Consider the reactions of the citric acid cycle. Which of the following statements regarding the cycle is *false*?
- A The outputs of the cycle are CO_2 , ATP and oxidized electron carriers ($\text{NADH} + \text{H}^+$ and FADH_2).
 - B The formation of citric acid from the products of pyruvate oxidation is the initial step in the cycle.
 - C Energy released during the cycle is captured in ATP, $\text{NADH} + \text{H}^+$ and FADH_2 .
 - D A certain percentage of energy is stored and subsequently lost in the form of CO_2 .
19. Which one of the following statements correctly describes cyclic photophosphorylation?
- A Cyclic photophosphorylation has both photosystems I and II.
 - B Cyclic photophosphorylation produces neither ATP nor $\text{NADPH} + \text{H}^+$.
 - C Electrons are cycled in cyclic photophosphorylation.
 - D Water is the ultimate source of electrons in cyclic photophosphorylation.
20. Which of the following statements about photorespiration is *false*?
- A Rubisco can function as both a carboxylase and an oxygenase.
 - B The carboxylase function of rubisco is favored at higher temperatures.
 - C The reaction of RuBP with O_2 forms a two-carbon compound called phosphoglycolate.
 - D The photorespiratory pathway appears to reverse photosynthesis because it releases CO_2 .

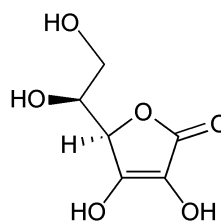
21. The diagram shows three routes through which substances can pass across a cell membrane.



Which correctly shows the routes for molecule 1 and molecule 2?



molecule 1



molecule 2

	molecule 1	molecule 2
A	Y	X
B	X	Z
C	X	Y
D	Z	Y

22. Some viruses are able to bind to the cell surface membrane before entering the host cell. Which sequence of events will lead to a virus invading a cell?
- A Binding to a cholesterol molecule, followed by endocytosis.
 - B Binding to a glycoprotein receptor, followed by exocytosis.
 - C Binding to a protein receptor, followed by endocytosis.
 - D Binding to the hydrophilic portion of a phospholipid, followed by exocytosis.

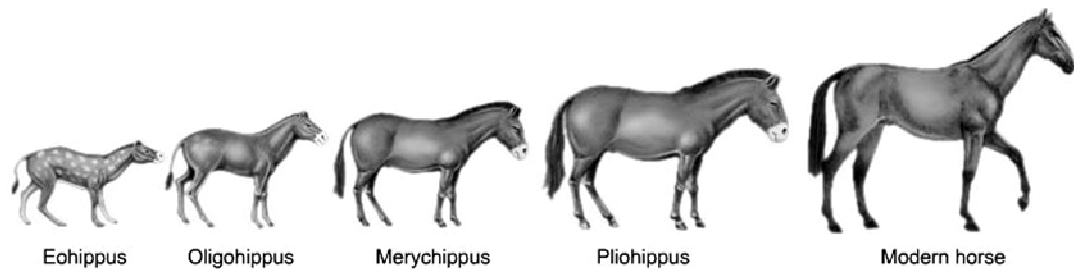
Use the following information for Questions 23 and 24.

The Norway rat (*Rattus norvegicus*), a widespread pest, was controlled for about a decade by the anticoagulant warfarin. This chemical substance inhibits the clotting of blood. Resistance to warfarin is governed by a dominant autosomal gene, *R*. The table below indicates the response of individuals to warfarin, vitamin K dependence and relative reproductive fitness. The highest fitness level is 1.00.

Characteristics	Genotypes of individuals		
	<i>rr</i>	<i>Rr</i>	<i>RR</i>
response to warfarin	susceptible	resistant	resistant
vitamin K dependence	no	intermediate	high
relative fitness	0.68	1.00	0.37

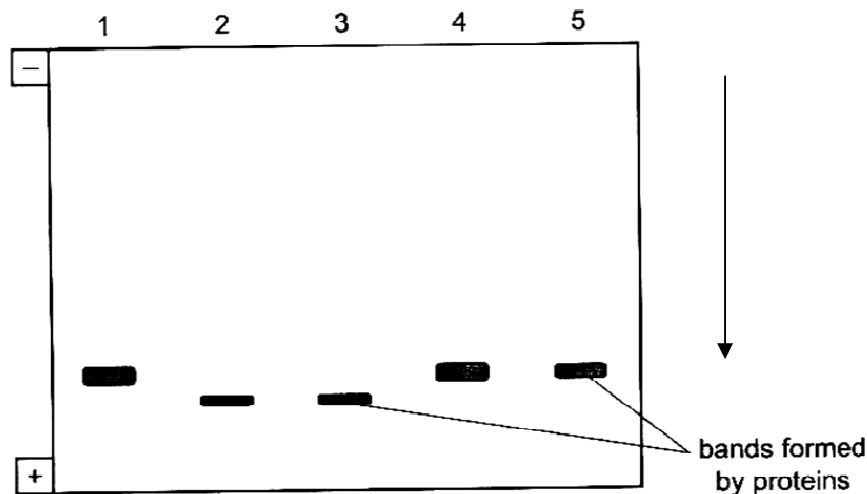
23. Which of the following is correct concerning the gene for resistance to warfarin?
- A It is a wild-type gene induced to mutate by warfarin.
 - B It is a mutation favoured by natural selection.
 - C It is increased in frequency by virtue of its dominant status.
 - D It offers an advantage to fitness levels under all conditions.
24. In a population where warfarin is no longer applied, the expectation is that the
- A heterozygotes will have a reproductive advantage over both homozygotes.
 - B frequency of the dominant allele will decline.
 - C dependence of the population on vitamin K will increase.
 - D proportion of recessive individuals will gradually decline.

25. The diagram below suggests the evolution of horses beginning from the Eohippus 58 million years ago.



Fossil records suggest that the ancestor of the modern horse is believed to have had relatively short legs. According to Darwinian views, the evolution of long-legged horses is best explained by

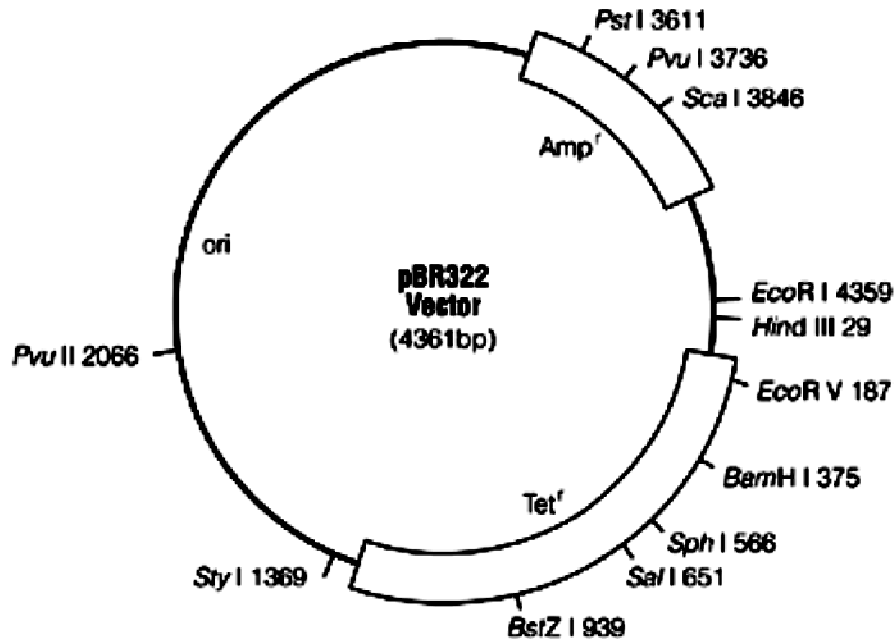
- A stabilising selection.
 - B directional selection.
 - C disruptive selection.
 - D acquired characteristics.
26. The diagram shows the results of electrophoresis of the same protein extracted from five different individuals.



A mutation occurred in individuals 2 and 3. Which one of the following can account for the different banding pattern?

- A A base has been deleted.
- B A base has been added.
- C A negatively charged amino acid replaced a non-charged amino acid.
- D A positively charged amino acid has been added to the protein.

27. The pBR322 vector is used to clone a eukaryotic gene, which has been digested by the restriction endonuclease *Bam*HI.



Following transformation, bacterial cells were grown in four different media, as shown below:

- I nutrient broth plus ampicillin
- II nutrient broth plus tetracycline
- III nutrient broth plus ampicillin and tetracycline
- IV nutrient broth without antibiotics

Which of the following media would bacterial cells that contain the recombinant plasmids grow in?

- A IV only
 - B I and II only
 - C II and III only
 - D I and IV only
28. Which of the following is a suitable source to locate stem cells?
- A Human blastocyst
 - B Umbilical cord blood
 - C Liver
 - D All of the above

- 29.** Why are somatic line therapies licensed for the treatment of genetic disorders rather than germ line therapies?
- A** Somatic cell manipulation is more effective than germ cell manipulation.
 - B** There is no control where the DNA inserts in germ line therapies.
 - C** The transformation is not carried on to the next generation in somatic line therapies.
 - D** Viruses can be used as vectors in somatic line therapies.
- 30.** Large quantities of useful products can be produced through genetic engineering involving
- 1** bacteria containing recombinant plasmids
 - 2** mammals producing substances in their milk
 - 3** transgenic plants
 - 4** microorganisms producing human hormones
 - 5** determination of number of genes in human
- A** 1,2,3,4 only
 - B** 2,3,4 only
 - C** 1,3,5 only
 - D** 2,4,5 only

ANSWERS

1	C	11	B	21	B
2	D	12	B	22	C
3	B	13	C	23	B
4	B	14	A	24	B
5	B	15	C	25	B
6	C	16	C	26	C
7	C	17	D	27	D
8	A	18	D	28	D
9	D	19	C	29	C
10	A	20	B	30	A